

Hand

Core idea

The hand is the distal, highly specialised part of the upper limb. It provides touch input, precision grip, power grip and fine manipulative movements. In exams, the highest-yield areas are palmar fascia, flexor retinaculum/carpal tunnel, intrinsic muscles, palmar arches, nerve distribution, palmar spaces and applied anatomy.

Main parts	Functional units	Key palmar structures	High-yield clinicals
Carpus, metacarpus and digits	Carpus, thumb, index finger and ulnar unit of middle-ring-little fingers	Palmar aponeurosis, flexor retinaculum, flexor sheaths, bursae, palmar arches	Carpal tunnel syndrome, Dupuytren contracture, trigger finger, space infections, scaphoid fracture

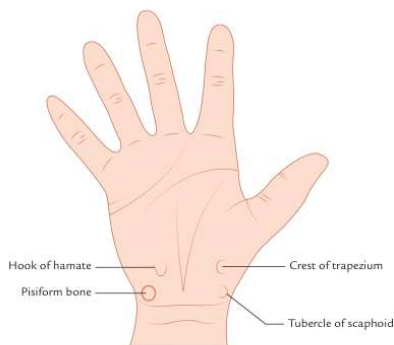


Fig. 11.1 Surface landmarks.

Source diagram: surface landmarks of the palm.

Surface landmarks to remember

- Scaphoid tubercle: felt at the base of thenar eminence, just lateral to tendon of flexor carpi radialis.
- Crest/tubercle of trapezium: felt deep and distal-lateral to the scaphoid tubercle.
- Pisiform: felt at the base of hypothenar eminence medially.
- Hook of hamate: felt about one finger breadth distal to pisiform.

1. SKIN, FASCIA AND PALMAR APONEUROSIS

The skin of the palm is thick, hairless and rich in sweat glands. It is closely anchored to the palmar aponeurosis, so it does not slide freely during grip. The palmar creases mark areas where skin is fixed to deep fascia and where repeated flexion occurs.

Skin of palm - exam points

- Thick and resistant to wear and tear.
- Rich in sweat glands but has no hair or sebaceous glands.

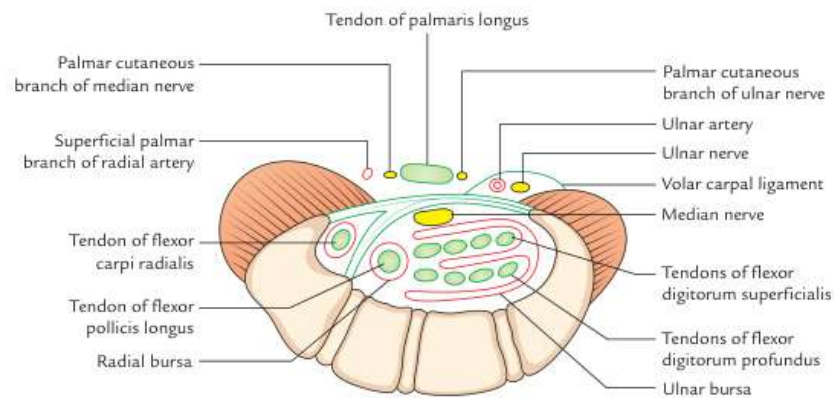
Superficial fascia

- Contains dense fibrous bands from skin to deep fascia.
- Contains superficial vessels, cutaneous

- Firmly fixed to underlying palmar aponeurosis.
 - Shows flexion creases due to firm fascial attachments.
- nerves and palmaris brevis.
 - Palmaris brevis wrinkles skin on the medial palm and improves grip.
 - Thick fat compartments cushion the hand during pressure.

Flexor retinaculum and carpal tunnel

The flexor retinaculum is a strong fibrous band bridging the anterior concavity of the carpus. Deep to it lies the carpal tunnel, containing the median nerve and nine long flexor tendons: four FDS, four FDP and one FPL. Flexor carpi radialis passes through a separate canal in the lateral part of the retinaculum.



Source diagram: transverse section of wrist through carpal tunnel.

Palmar aponeurosis

Feature	Important point
Apex	Directed proximally; continuous with the tendon of palmaris longus and flexor retinaculum.
Base	Directed distally; divides into four slips for medial four digits and blends with fibrous flexor sheaths.
Function	Fixes skin for grip and protects underlying tendons, vessels and nerves.

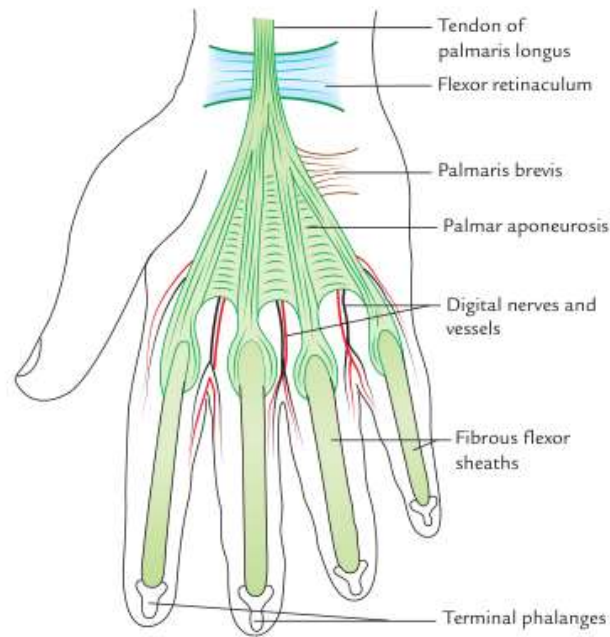


Fig. 11.6 Palmar aponeurosis.

Source diagram: palmar aponeurosis.

Clinical correlation - Dupuytren contracture

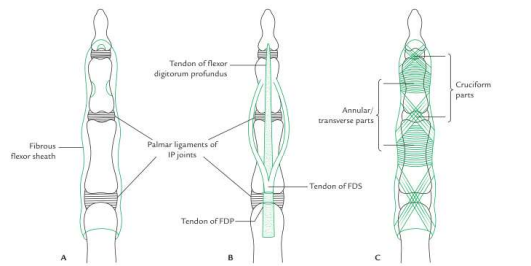
Dupuytren contracture is progressive fibrous thickening of palmar aponeurosis, usually affecting the medial palm. It produces flexion deformity of ring and little fingers. Fasciectomy may be needed when function is severely impaired.

2. FLEXOR SHEATHS, BURSAE AND TENDON TUNNELS

Fibrous flexor sheaths are deep fascial thickenings over the anterior surface of digits. They form osseofibrous tunnels with phalanges and palmar ligaments, keeping the long flexor tendons close to the bones during finger flexion.

Structure	Arrangement	Function / clinical note
Fibrous flexor sheath	Annular fibres in front of phalanges and cruciate fibres in front of joints.	Prevents bow-stringing of tendons during flexion.
Digital synovial sheath	Double-layered synovial sheath around flexor tendons within digits.	Lubricates tendons as they move through osseofibrous tunnels.
Trigger finger	Local thickening of long flexor tendon or sheath catches in tunnel.	Finger locks in flexion and releases with a sudden click.

hes over the long flexor tendon to form the *fibrous* continuous with the distal slips of the palmar aponeurosis.



.8 Fibrous flexor sheaths of the fingers: A, attachment of the sheath; B, tendons passing through the sheath; C, arrangement of fibres within sheath—cruciate fibres in front of joints and transverse fibres in front of bones (FDS = flexor digitorum superficialis, FDP = flexor digitorum profundus, FPL = flexor pollicis longus).

Source diagram: fibrous flexor sheaths.

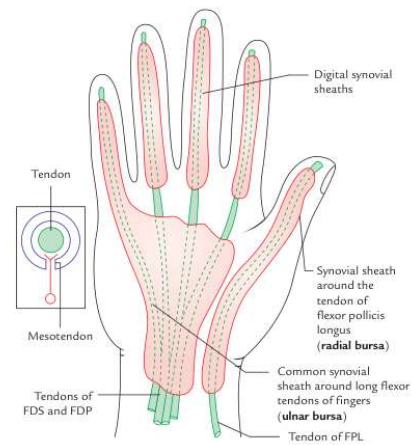


Fig. 11.9 Synovial sheaths around the long flexor tendons. Figure in the inset shows two layers of synovial sheath and mesotendon (FDS = flexor digitorum superficialis, FDP = flexor digitorum profundus, FPL = flexor pollicis longus).

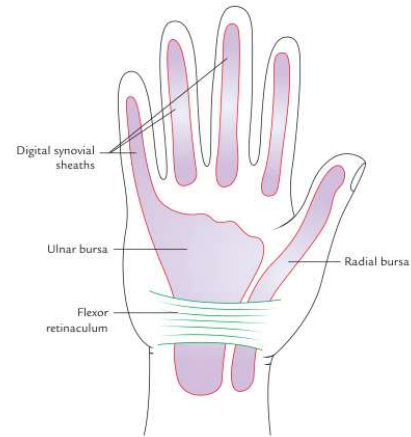


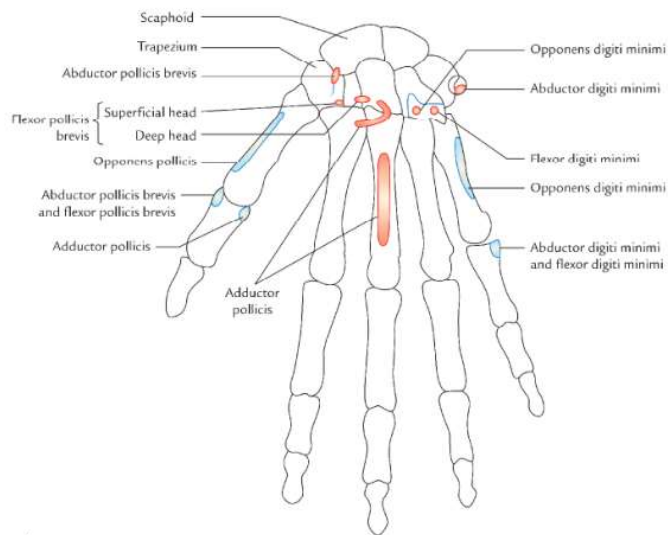
Fig. 11.10 Ulnar bursa, radial bursa, and digital synovial sheaths.

Source diagram: ulnar bursa, radial bursa and digital sheaths.

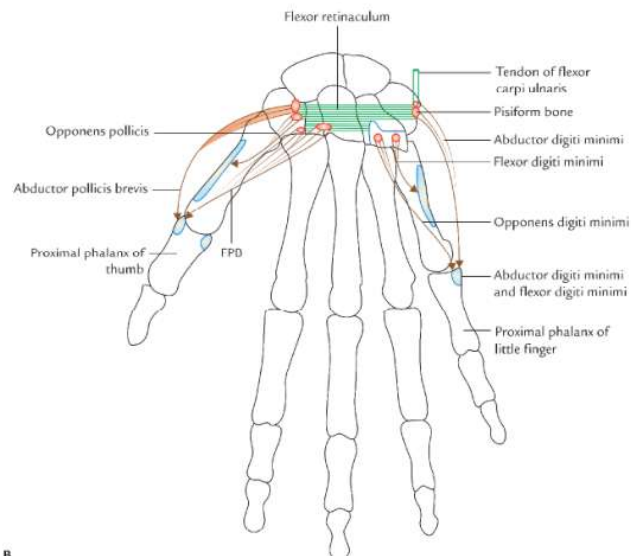
Bursa / sheath	Contains / extent	Clinical importance
Ulnar bursa	Common synovial sheath for tendons of FDS and FDP. It extends into forearm proximal to flexor retinaculum and into palm.	Often communicates with digital sheath of little finger; infection can spread to forearm.
Radial bursa	Synovial sheath around tendon of flexor pollicis longus.	May communicate with digital sheath of thumb.
Digital sheaths	Synovial sheaths for flexor tendons within digits.	Index, middle and ring sheaths are usually separate from ulnar bursa.

3. INTRINSIC MUSCLES OF THE HAND

Intrinsic muscles begin and end within the hand. They refine the actions of long flexors and extensors, especially for precision grip, thumb opposition and coordinated finger movements.



A



B

Source diagram: thenar and hypothenar muscles with bony attachments.

Group	Muscles	Nerve supply	Action / importance
Thenar	Abductor pollicis brevis, flexor pollicis brevis, opponens pollicis	Recurrent branch of median nerve; deep head of FPD may receive ulnar supply.	Forms thenar eminence; chief muscles for opposition of thumb.
Hypothenar	Abductor digiti minimi, flexor digiti minimi, opponens digiti minimi	Deep branch of ulnar nerve.	Forms hypothenar eminence; moves little finger for power grip.
Adductor pollicis	Oblique head and transverse head	Deep branch of ulnar nerve.	Adducts thumb; gives power to grip; tested by Froment sign.
Lumbricals	Four slender muscles arising from FDP	Lateral two: median nerve. Medial two:	Flex MP joints and extend IP joints

Group	Muscles	Nerve supply	Action / importance
	tendons	deep branch of ulnar nerve.	through extensor expansion.
Interossei	Four dorsal and three palmar interossei	Deep branch of ulnar nerve.	Dorsal abduct digits; palmar adduct digits. PAD and DAB mnemonic.

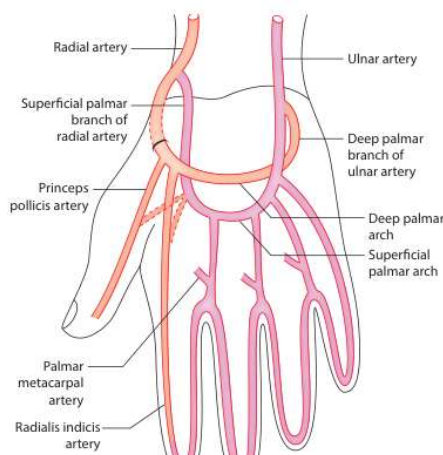
Ulnar nerve memory point

The ulnar nerve supplies almost all intrinsic muscles of the hand except the thenar muscles and lateral two lumbricals. Because these muscles control fine movements, the ulnar nerve is often called the musician's nerve.

4. BLOOD SUPPLY OF THE HAND

The palm is supplied mainly by superficial and deep palmar arterial arches. These arches allow the hand to maintain blood flow during grasping and changes in position.

Feature	Superficial palmar arch	Deep palmar arch
Formation	Mainly continuation of ulnar artery, completed by superficial palmar branch of radial artery.	Mainly continuation of radial artery, completed by deep palmar branch of ulnar artery.
Position	Superficial to long flexor tendons; lies across centre of palm near distal border of fully extended thumb.	Deep to long flexor tendons; about 1 cm proximal to superficial arch.
Branches	Three common palmar digital arteries and one proper palmar digital artery.	Palmar metacarpal arteries, perforating arteries and recurrent branches.



Clinical note - bleeding from palmar arches

Laceration of palmar arterial arches can produce profuse bleeding. Compression of the brachial artery against the humerus is an effective proximal control method while definitive management is planned.

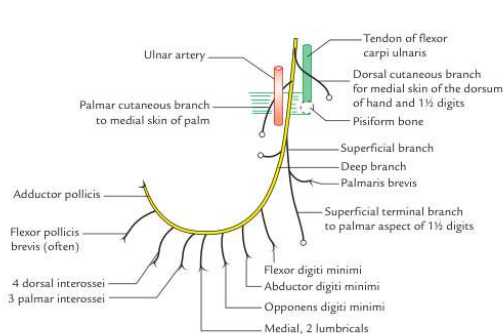
- Avoid blind clamping inside the palm because digital nerves and tendons are very close to the arteries.
- Bleeding is brisk because palmar arches have rich anastomoses.
- Check capillary refill and digital sensation after deep palmar wounds.

Source diagram: superficial and deep palmar arterial arches.

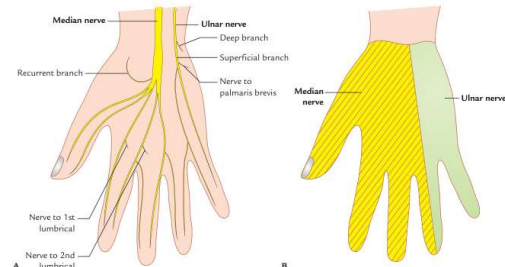
5. NERVE SUPPLY AND SENSORY TERRITORIES

The hand receives important branches from median, ulnar and radial nerves. For practical anatomy, divide the palm into median territory laterally, ulnar territory medially and radial sensory territory mainly on dorsolateral hand.

Nerve	Motor supply in hand	Sensory supply
Median nerve	Thenar muscles through recurrent branch and lateral two lumbricals.	Palmar skin of lateral 3½ digits and adjacent palm; also nail beds/distal dorsal parts of same digits.
Ulnar nerve	Hypothenar muscles, adductor pollicis, palmar and dorsal interossei, medial two lumbricals and palmaris brevis.	Medial 1½ digits and adjacent medial palm/dorsum.
Radial nerve	No intrinsic motor supply in hand.	Dorsolateral hand and proximal dorsal parts of lateral 3½ digits, except nail beds.
Clinical focus	Median compression causes thenar weakness; ulnar injury causes loss of fine intrinsic movements.	Sensory loss pattern helps localize lesion. Always compare both hands.



Source diagram: course and distribution of ulnar nerve in hand.



Source diagram: median and ulnar nerves in palm and sensory territories.

High-yield nerve lesions

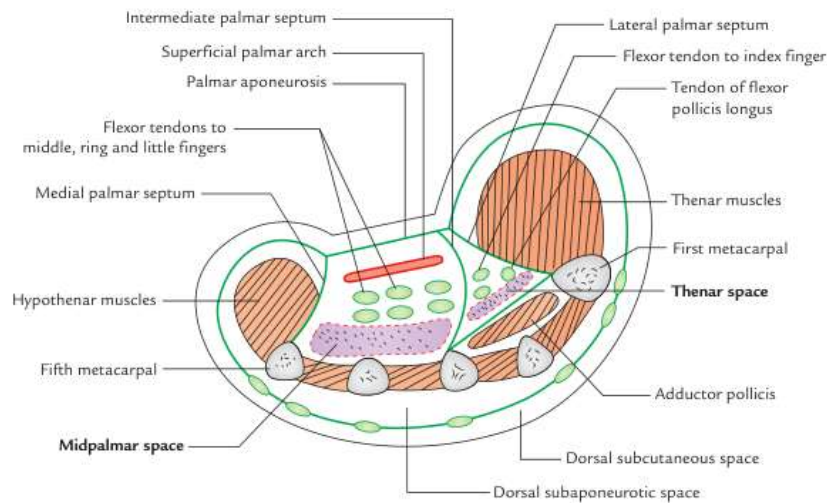
Carpal tunnel syndrome compresses the median nerve deep to flexor retinaculum. Guyon canal syndrome compresses the ulnar nerve at the wrist and causes sensory loss in medial 1½ fingers with weakness of intrinsic muscles.

6. FASCIAL SPACES OF THE HAND

Fascial spaces are normally potential spaces filled with loose connective tissue. They become clinically important when infected fluid or pus collects in them. The most important palmar spaces are the midpalmar space and thenar space.

Clinical logic of hand infections

Palmar spaces are closed by septa, so pus tracks through predictable channels such as lumbrical canals, bursae and web spaces. Dorsal swelling may be marked because the dorsal fascia is loose, even when the primary infection began in the palm.



Source diagram: cross-section of hand showing palmar and dorsal spaces.

Space	Boundaries / position	Clinical spread
Midpalmar space	Located under medial half of hollow of palm; deep to palmar aponeurosis, superficial palmar arch, digital nerves/vessels, ulnar bursa and medial lumbricals.	Communicates proximally with forearm space of Parona and distally with medial three web spaces via lumbrical canals.
Thenar space	Located under lateral half of hollow of palm; related to radial bursa, index flexor tendons and first lumbrical.	Communicates distally with first web space; infection causes swelling of thenar region.
Pulp space	Subcutaneous fibrofatty space at the fingertip, divided by fibrous septa.	Infection causes throbbing pain and may compromise vessels to distal phalanx.
Web space	Subcutaneous space in interdigital cleft containing lumbrical tendon, interosseous tendon, digital nerve and vessels.	Important route for spread from palmar spaces to web.

7. DORSUM OF HAND AND ANATOMICAL SNUFFBOX

The dorsum of the hand has loose skin and superficial veins. Swelling therefore becomes prominent on the dorsum even when the primary infection is palmar. The anatomical snuffbox is an important triangular depression on the lateral dorsum of the wrist when the thumb is extended.

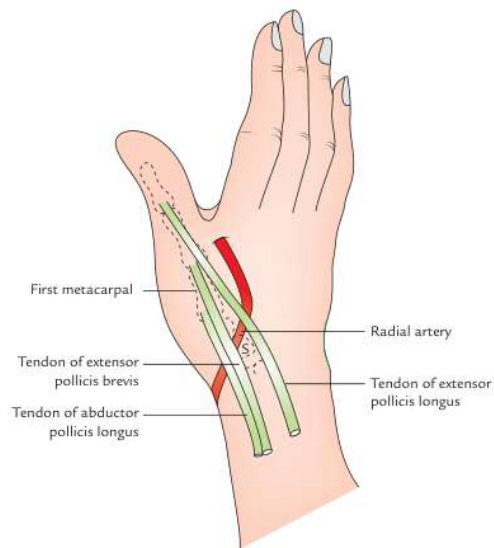


Fig. 11.28 Boundaries and contents of the anatomical snuffbox (S = scaphoid).

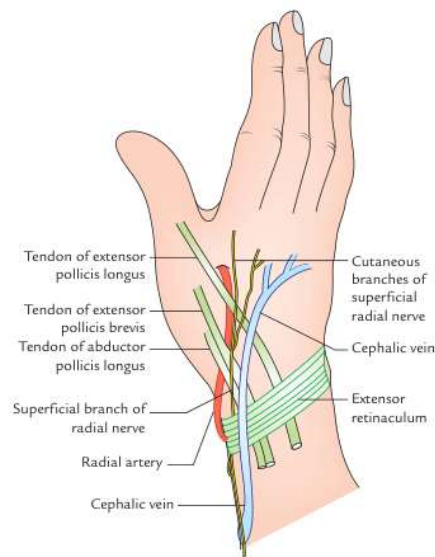


Fig. 11.29 Structures crossing the roof of anatomical snuffbox.

Source diagrams: anatomical snuffbox boundaries and structures crossing its roof.

Item	Details
Anterolateral boundary	Tendons of abductor pollicis longus and extensor pollicis brevis.
Posteromedial boundary	Tendon of extensor pollicis longus.
Floor	Scaphoid and trapezium.
Contents / roof	Radial artery is in the floor. Cephalic vein and superficial radial nerve branches cross the roof.

Clinical significance of snuffbox

Pulsation of radial artery can be felt in the anatomical snuffbox. Tenderness in the snuffbox after a fall strongly suggests scaphoid fracture. The cephalic vein at this site is also used for intravenous access.

8. JOINTS OF WRIST, HAND AND FINGERS

The joints of the hand are arranged to provide both stability and highly mobile precision movements. The wrist joint gives broad positioning of the hand, while carpometacarpal, metacarpophalangeal and interphalangeal joints refine grip and manipulation.

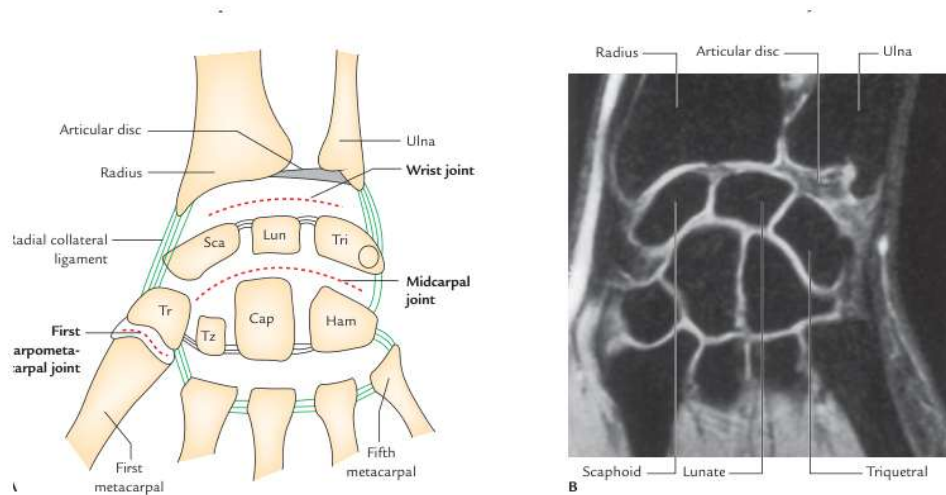
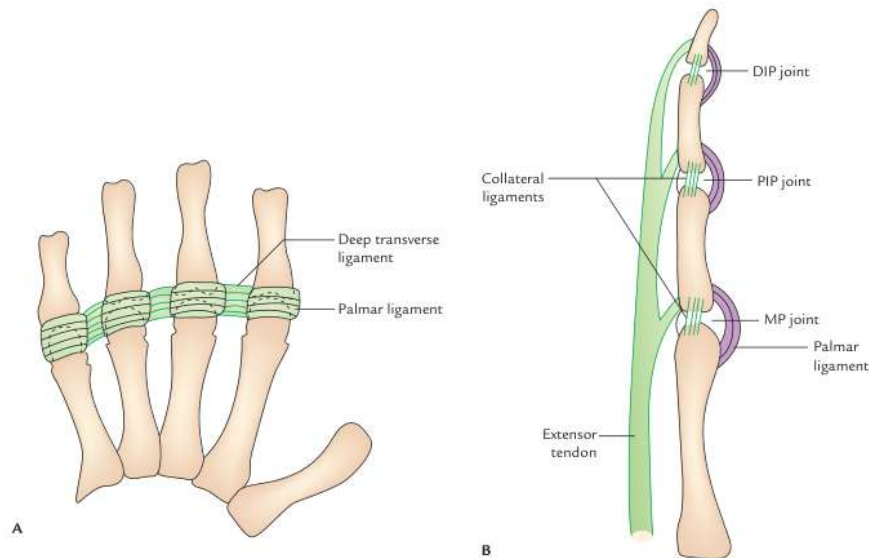


Fig. 12.1 Coronal section through wrist region: **A**, schematic diagram; **B**, as seen in magnetic resonance imaging, showing wrist joint, midcarpal joint, intercarpal joints, carpometacarpal joints. (Source **B**: Fig. 7.91C, Page 710, *Gray's Anatomy for students*. Richard I Drake, Wayne Vogl, Adam WM Mitchell. Copyright Elsevier Inc. 2005. All rights reserved.)

Source diagram: coronal section through wrist showing wrist, midcarpal, intercarpal and carpometacarpal joints.

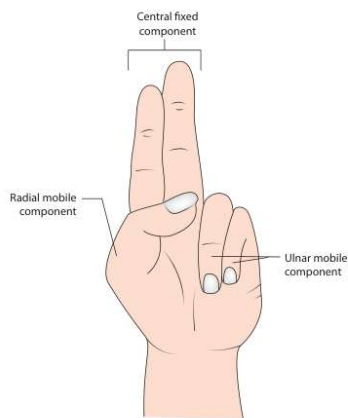
Joint	Type / articular surfaces	Movements / notes
Wrist joint	Synovial ellipsoid joint between distal radius + articular disc and proximal row of carpal bones except pisiform.	Flexion, extension, abduction, adduction and circumduction.
Intercarpal joints	Plane synovial joints between carpal bones.	Gliding movements; midcarpal joint assists wrist movements.
CMC joints 2-5	Plane synovial joints; second and third are relatively fixed.	Stable base for palm; fourth and fifth are more mobile for cupping.
First CMC joint	Saddle synovial joint between trapezium and first metacarpal.	Flexion, extension, abduction, adduction, opposition and circumduction of thumb.
MP and IP joints	MP: ellipsoid/condylar. IP: hinge synovial joints.	MP joints allow flexion-extension, abduction-adduction and circumduction. IP joints allow flexion-extension.



12.5 Joints of the fingers: **A**, MP joints showing palmar and deep transverse ligaments; **B**, MP, PIP, and DIP joints showing palmar and collateral ligaments (DIP = distal interphalangeal, PIP = proximal interphalangeal, MP = metacarpophalangeal).

Source diagram: MP, PIP and DIP joints with ligaments.

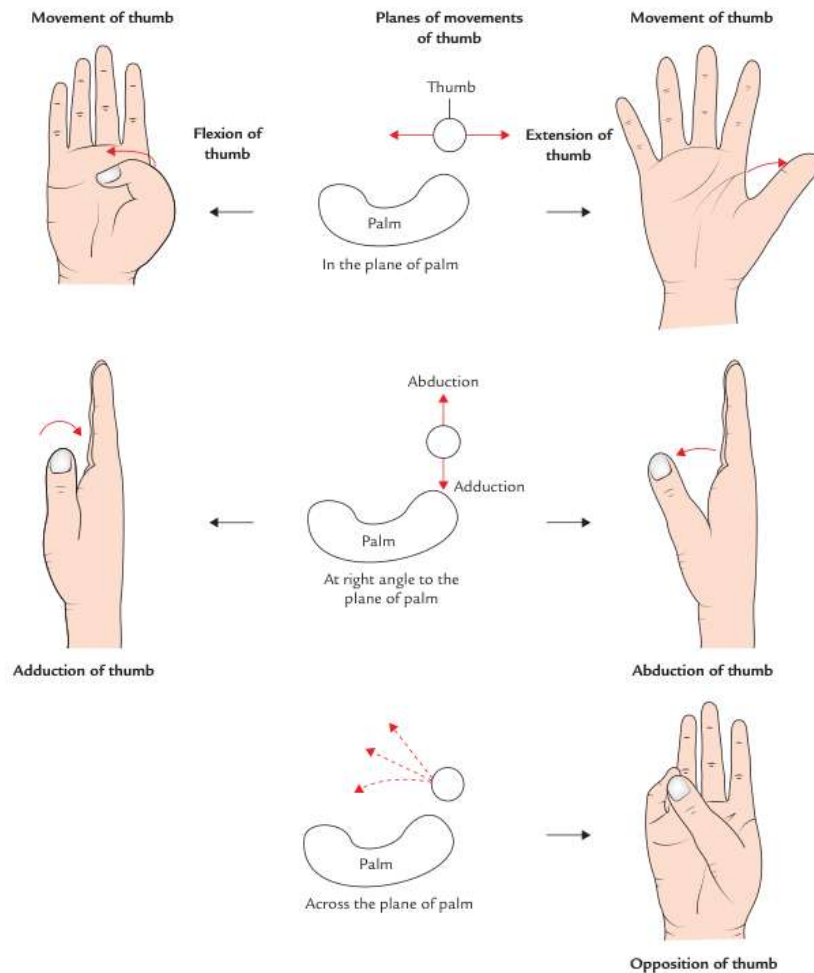
9. FUNCTIONAL COMPONENTS AND MOVEMENTS



Source diagram: functional components of the hand.

Functional components

- Central fixed component: metacarpals of index and middle fingers form the stable central backbone.
- Radial mobile component: thumb; important for precision manipulation and opposition.
- Ulnar mobile component: ring and little fingers; important for power grip and hook grip.
- Radial digital tripod: thumb, index and middle fingers act together in precision grip.
- Power grip depends on stable carpus, flexor tendons and mobile ulnar digits.
- Precision grip depends on thumb opposition against the index and middle fingers.
- The arches of the hand deepen the hollow of palm and make grasping more secure.
- Loss of thumb is functionally severe because it affects both power and precision grip.



12.11 Movements of the thumb.

Source diagram: movements of the thumb.

Movement	Thumb	Fingers
Flexion / extension	Occurs in plane of palm; thumb flexion crosses the palm.	Occurs at MP, PIP and DIP joints.
Abduction / adduction	Occurs at right angle to plane of palm.	Occurs at MP joints relative to the imaginary axis of middle finger.
Opposition	Thumb touches tips of fingers by combined flexion, abduction and medial rotation.	No true opposition except functional approximation with thumb.
Circumduction	Possible due to first CMC mobility.	Possible mainly at MP joints.

10. APPLIED ANATOMY - QUICK CLINICAL REVISION

Condition	Anatomical basis	Clinical clue
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Condition	Anatomical basis	Clinical clue
Carpal tunnel syndrome	Median nerve compressed beneath flexor retinaculum along with flexor tendons.	Pain/paraesthesia in lateral 3½ digits; thenar weakness in severe cases.
Dupuytren contracture	Fibrosis and shortening of palmar aponeurosis.	Flexion deformity of ring and little fingers.
Trigger finger	Thickened tendon/sheath catches inside fibrous flexor sheath.	Finger locks then releases with a click.
Space infection	Pus spreads along midpalmar/thenar spaces, lumbrical canals and bursae.	Dorsal swelling may appear due to loose dorsal fascia.
Scaphoid fracture	Scaphoid forms floor of anatomical snuffbox.	Snuffbox tenderness after fall on outstretched hand.
Ulnar canal syndrome	Ulnar nerve compressed in Guyon canal near pisiform and hook of hamate.	Hypoesthesia of medial 1½ fingers and intrinsic muscle weakness.
Froment sign	Adductor pollicis paralysis due to ulnar nerve lesion.	Thumb IP flexes while holding paper due to compensatory FPL action.

11. LAST-MINUTE EXAM SHEET

Must write headings	Most tested facts	Clinical keywords
• Skin and fascia • Flexor retinaculum/carpal tunnel • Palmar aponeurosis • Bursae and synovial sheaths • Intrinsic muscles • Nerves and blood supply • Fascial spaces • Snuffbox and applied	• Median nerve: thenar + lateral two lumbricals • Ulnar nerve: most intrinsic muscles • Superficial arch mainly ulnar artery • Deep arch mainly radial artery • Dorsal interossei abduct; palmar interossei adduct	• Carpal tunnel syndrome • Dupuytren contracture • Trigger finger • Thenar/midpalmar space infection • Guyon canal syndrome • Scaphoid tenderness in snuffbox

One-line conclusion

The hand is designed for both stability and dexterity: palmar fascia and arches protect and support it, intrinsic muscles create precision, and the median/ulnar/radial nerves provide the motor-sensory pattern needed for skilled function.

Long-answer order	Memory hooks
• Definition and parts of hand • Skin and superficial fascia	• LOAF = median nerve muscles: lateral two lumbricals, opponens pollicis,

• Flexor retinaculum and carpal tunnel • Palmar aponeurosis • Flexor sheaths, bursae and spaces • Intrinsic muscles • Palmar arches and nerve supply • Applied anatomy	abductor pollicis brevis, flexor pollicis brevis. • PAD = palmar interossei adduct; DAB = dorsal interossei abduct. • Superficial palmar arch is mainly ulnar; deep palmar arch is mainly radial. • Snuffbox tenderness means think scaphoid fracture.
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Topic	Must name	Must add	Clinical link
Palm	Palmar aponeurosis	Median + ulnar digital branches	Dupuytren contracture
Tendons	Ulnar/radial bursae	Fibrous flexor sheaths	Trigger finger + spread of infection